Calibration Plots for Uncertainty Metric Frequency of Answer

gpt-oss-20b

ECE	0.0178
AUROC	0.8261
Invalid Answers	2/10000
Invalid Metric Scores	0/9998
Norm. Entropy of Bucket Counts	0.1567
Accuracy	0.9378

gpt-oss-120b

ECE	0.0187
AUROC	0.7652
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1273
Accuracy	0.9684

Ministral-8B

ECE	0.0587
AUROC	0.7622
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.8206
Accuracy	0.7218

Mistral-Nemo-7B

ECE	0.0635
AUROC	0.7861
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7820
Accuracy	0.7471

Mistral-Small-3.2-24B

ECE	0.0240
AUROC	0.8536
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3768
Accuracy	0.9210

Llama-3.3-70B

ECE	0.0401
AUROC	0.8138
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2907
Accuracy	0.9196

Llama-4-Scout-17B

ECE	0.0336
AUROC	0.7929
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2282
Accuracy	0.9360

Qwen3-30B-A3B-Instruct

ECE	0.0198
AUROC	0.7323
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0932
Accuracy	0.9710

Qwen3-30B-A3B-Thinking

ECE	0.7500
AUROC	0.5000
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0000
Accuracy	0.2500

DeepSeek-Llama-70B

ECE	0.0440
AUROC	0.8493
Invalid Answers	0/10000
Invalid Metric Scores	0/9982
Norm. Entropy of Bucket Counts	0.3884
Accuracy	0.8648

DeepSeek-Qwen-32B

ECE	0.0553
AUROC	0.8105
Invalid Answers	29/10000
Invalid Metric Scores	0/9971
Norm. Entropy of Bucket Counts	0.3454
Accuracy	0.8776

google/gemma-3-27b

ECE	0.0623
AUROC	0.7823
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3743
Accuracy	0.8861

ARC-Easy

ECE	0.0067
AUROC	0.8857
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0963
Accuracy	0.9862

ECE	0.0064
AUROC	0.8441
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0372
Accuracy	0.9926

ECE	0.0246
AUROC	0.8798
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3094
Accuracy	0.9558

ECE	0.0219
AUROC	0.8994
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4291
Accuracy	0.9181

ECE	0.0091
AUROC	0.9050
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1440
Accuracy	0.9746

ECE	0.0120
AUROC	0.8795
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0726
Accuracy	0.9824

ECE	0.0076
AUROC	0.7897
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0637
Accuracy	0.9878

ECE	0.0106
AUROC	0.7697
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0598
Accuracy	0.9842

ECE	0.7500
AUROC	0.5000
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0000
Accuracy	0.2500

ECE	0.0076
AUROC	0.9288
Invalid Answers	0/10000
Invalid Metric Scores	0/9999
Norm. Entropy of Bucket Counts	0.1877
Accuracy	0.9748

ECE	0.0114
AUROC	0.9128
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2344
Accuracy	0.9633

ECE	0.0171
AUROC	0.8228
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1049
Accuracy	0.9732

ARC-Challenge

ECE	0.0078
AUROC	0.9790
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1377
Accuracy	0.9773

ECE	0.0065
AUROC	0.8716
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0784
Accuracy	0.9878

ECE	0.0197
AUROC	0.8706
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6496
Accuracy	0.8405

ECE	0.0348
AUROC	0.8675
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7315
Accuracy	0.8586

ECE	0.0181
AUROC	0.8841
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2381
Accuracy	0.9510

ECE	0.0109
AUROC	0.7146
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1053
Accuracy	0.9691

ECE	0.0188
AUROC	0.7630
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0973
Accuracy	0.9713

ECE	0.0200
AUROC	0.7330
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0879
Accuracy	0.9729

ECE	0.7500
AUROC	0.5000
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0000
Accuracy	0.2500

ECE	0.0233
AUROC	0.8855
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3106
Accuracy	0.9394

ECE	0.0159
AUROC	0.8768
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3450
Accuracy	0.9277

ECE	0.0413
AUROC	0.8454
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1501
Accuracy	0.9445

SciQ

ECE	0.0187
AUROC	0.7667
Invalid Answers	1/10000
Invalid Metric Scores	0/9999
Norm. Entropy of Bucket Counts	0.1326
Accuracy	0.9658

ECE	0.0152
AUROC	0.8015
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0916
Accuracy	0.9776

ECE	0.0217
AUROC	0.8015
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5283
Accuracy	0.8899

ECE	0.0193
AUROC	0.8729
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4334
Accuracy	0.9051

ECE	0.0201
AUROC	0.8113
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2109
Accuracy	0.9547

ECE	0.0145
AUROC	0.7587
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1001
Accuracy	0.9782

ECE	0.0138
AUROC	0.8486
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1162
Accuracy	0.9732

ECE	0.0231
AUROC	0.7583
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1082
Accuracy	0.9674

ECE	0.7499
AUROC	0.5004
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.0034
Accuracy	0.2501

ECE	0.0147
AUROC	0.8247
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1703
Accuracy	0.9656

ECE	0.0252
AUROC	0.8490
Invalid Answers	2/10000
Invalid Metric Scores	0/9998
Norm. Entropy of Bucket Counts	0.2931
Accuracy	0.9520

ECE	0.0371
AUROC	0.7742
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1420
Accuracy	0.9501

GPQA

ECE	0.0722
AUROC	0.7931
Invalid Answers	266/10000
Invalid Metric Scores	0/9734
Norm. Entropy of Bucket Counts	0.6526
Accuracy	0.7995

ECE	0.0749
AUROC	0.7925
Invalid Answers	0/10000
Invalid Metric Scores	0/9994
Norm. Entropy of Bucket Counts	0.5356
Accuracy	0.8193

ECE	0.2100
AUROC	0.5206
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.9060
Accuracy	0.5123

ECE	0.1900
AUROC	0.5925
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.8978
Accuracy	0.5649

ECE	0.1508
AUROC	0.6685
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7923
Accuracy	0.6680

ECE	0.1446
AUROC	0.6769
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6447
Accuracy	0.6999

ECE	0.1706
AUROC	0.6660
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5980
Accuracy	0.7234

ECE	0.1156
AUROC	0.7384
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5955
Accuracy	0.7731

ECE	0.7282
AUROC	0.5094
Invalid Answers	1243/10000
Invalid Metric Scores	0/9757
Norm. Entropy of Bucket Counts	0.1450
Accuracy	0.2513

ECE	0.1498
AUROC	0.8719
Invalid Answers	78/9000
Invalid Metric Scores	0/9922
Norm. Entropy of Bucket Counts	0.4188
Accuracy	0.8397

ECE	0.2058
AUROC	0.6035
Invalid Answers	830/10000
Invalid Metric Scores	0/9570
Norm. Entropy of Bucket Counts	0.8208
Accuracy	0.6010

ECE	0.2169
AUROC	0.6135
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7408
Accuracy	0.6303

GSM8K

ECE	0.0342
AUROC	0.7167
Invalid Answers	4/2500
Invalid Metric Scores	0/2496
Norm. Entropy of Bucket Counts	0.1878
Accuracy	0.9459

ECE	0.0433
AUROC	0.6421
Invalid Answers	1/2500
Invalid Metric Scores	0/2498
Norm. Entropy of Bucket Counts	0.0937
Accuracy	0.9484

ECE	0.0288
AUROC	0.9416
Invalid Answers	0/2500
Invalid Metric Scores	0/2492
Norm. Entropy of Bucket Counts	0.5074
Accuracy	0.8973

ECE	0.0640
AUROC	0.9026
Invalid Answers	1/2500
Invalid Metric Scores	0/2488
Norm. Entropy of Bucket Counts	0.4124
Accuracy	0.8549

ECE	0.0338
AUROC	0.7214
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.1189
Accuracy	0.9544

ECE	0.0248
AUROC	0.6268
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.0539
Accuracy	0.9728

ECE	0.0313
AUROC	0.7005
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.0627
Accuracy	0.9660

ECE	0.0380
AUROC	0.6563
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.0674
Accuracy	0.9584

ECE	0.0494
AUROC	0.6295
Invalid Answers	134/2500
Invalid Metric Scores	0/2366
Norm. Entropy of Bucket Counts	0.3518
Accuracy	0.9611

ECE	0.0356
AUROC	0.8197
Invalid Answers	1/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.2833
Accuracy	0.9492

ECE	0.0371
AUROC	0.7719
Invalid Answers	1/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.2448
Accuracy	0.9373

ECE	0.0474
AUROC	0.6389
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.1285
Accuracy	0.9424

SVamp

ECE	0.0351
AUROC	0.6651
Invalid Answers	4/2500
Invalid Metric Scores	0/2496
Norm. Entropy of Bucket Counts	0.1319
Accuracy	0.9539

ECE	0.0317
AUROC	0.7098
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.1253
Accuracy	0.9372

ECE	0.0319
AUROC	0.8243
Invalid Answers	54/2500
Invalid Metric Scores	0/2441
Norm. Entropy of Bucket Counts	0.3754
Accuracy	0.9309

ECE	0.0325
AUROC	0.7039
Invalid Answers	39/2500
Invalid Metric Scores	0/2461
Norm. Entropy of Bucket Counts	0.1405
Accuracy	0.9065

ECE	0.0394
AUROC	0.7819
Invalid Answers	9/2500
Invalid Metric Scores	0/2491
Norm. Entropy of Bucket Counts	0.1892
Accuracy	0.9398

ECE	0.0275
AUROC	0.8330
Invalid Answers	8/2500
Invalid Metric Scores	0/2492
Norm. Entropy of Bucket Counts	0.1400
Accuracy	0.9570

ECE	0.0327
AUROC	0.8354
Invalid Answers	1/2500
Invalid Metric Scores	0/2499
Norm. Entropy of Bucket Counts	0.1692
Accuracy	0.9400

ECE	0.0298
AUROC	0.6493
Invalid Answers	14/2500
Invalid Metric Scores	0/2486
Norm. Entropy of Bucket Counts	0.0764
Accuracy	0.9646

ECE	0.0349
AUROC	0.6039
Invalid Answers	55/2500
Invalid Metric Scores	0/2445
Norm. Entropy of Bucket Counts	0.2396
Accuracy	0.9644

ECE	0.0535
AUROC	0.9358
Invalid Answers	201/2500
Invalid Metric Scores	0/2299
Norm. Entropy of Bucket Counts	0.5448
Accuracy	0.9317

ECE	0.0381
AUROC	0.9062
Invalid Answers	24/2500
Invalid Metric Scores	0/2476
Norm. Entropy of Bucket Counts	0.2461
Accuracy	0.9487

ECE	0.0318
AUROC	0.7610
Invalid Answers	1/2500
Invalid Metric Scores	0/2499
Norm. Entropy of Bucket Counts	0.1103
Accuracy	0.9560

SciBench

ECE	0.1004
AUROC	0.8321
Invalid Answers	452/500
Invalid Metric Scores	0/2455
Norm. Entropy of Bucket Counts	0.5115
Accuracy	0.4134

ECE	0.2128
AUROC	0.8411
Invalid Answers	9/2500
Invalid Metric Scores	0/2491
Norm. Entropy of Bucket Counts	0.9198
Accuracy	0.4255

ECE	0.0837
AUROC	0.9075
Invalid Answers	55/2500
Invalid Metric Scores	0/2443
Norm. Entropy of Bucket Counts	0.6565
Accuracy	0.1820

ECE	0.0891
AUROC	0.8960
Invalid Answers	69/2500
Invalid Metric Scores	0/2440
Norm. Entropy of Bucket Counts	0.5564
Accuracy	0.1799

ECE	0.1269
AUROC	0.8565
Invalid Answers	11/2500
Invalid Metric Scores	0/2489
Norm. Entropy of Bucket Counts	0.9807
Accuracy	0.5195

ECE	0.1311
AUROC	0.8465
Invalid Answers	14/2500
Invalid Metric Scores	0/2486
Norm. Entropy of Bucket Counts	0.8737
Accuracy	0.4610

ECE	0.1643
AUROC	0.7984
Invalid Answers	9/2500
Invalid Metric Scores	0/2491
Norm. Entropy of Bucket Counts	0.7487
Accuracy	0.6078

ECE	0.1865
AUROC	0.7686
Invalid Answers	122/2500
Invalid Metric Scores	0/2488
Norm. Entropy of Bucket Counts	0.6219
Accuracy	0.6455

ECE	0.1353
AUROC	0.8016
Invalid Answers	743/2500
Invalid Metric Scores	0/2757
Norm. Entropy of Bucket Counts	0.8028
Accuracy	0.5913

ECE	0.1232
AUROC	0.8121
Invalid Answers	328/2500
Invalid Metric Scores	0/2172
Norm. Entropy of Bucket Counts	0.9012
Accuracy	0.4337

ECE	0.1482
AUROC	0.8270
Invalid Answers	291/2500
Invalid Metric Scores	0/2209
Norm. Entropy of Bucket Counts	0.9503
Accuracy	0.4301

ECE	0.1231
AUROC	0.8289
Invalid Answers	6/2500
Invalid Metric Scores	0/2494
Norm. Entropy of Bucket Counts	0.8477
Accuracy	0.5485